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Pump with variable suction/discharge amount and drive device composed of the pump and driving method thereof

Abstract

A pump with variable suction/discharge amount and a transmission drive device and a driving method thereof. The pump is a rotary vane pump having a vane chamber body. The vane chamber body is composed of a fixed wall member, a movable wall member, a movable vane chamber sleeve and a vane rotor. The vane chamber is extendable/retractable in an axial direction of the vane rotor. At least two pumps are assembled in communication with each other to form a closed loop for the active pump to drive the passive pump. In operation, passive the vane chambers of the active pump and the passive pump are automatically extended/retracted and modulated until the driving force and the load resistance achieve a balanced state passive, the vane chambers and the rotational speeds of the active pump and the passive pump are automatically adjusted to be in inverse proportion to each other.

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Background/Summary

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The present invention relates generally to a pump with variable suction/discharge amount and a drive device composed of the pump and a driving method thereof, and more particularly to a rotary vane pump composed of a fixed wall member, a movable wall member, a movable vane chamber sleeve and a vane rotor. The rotary vane pump has a vane chamber, which is extendable/retractable in an axial direction of the vane rotor. Accordingly, the capacity of the vane chamber is variable to form the pump with variable suction/discharge amount. In addition, at least two pumps with variable suction/discharge amount can be assembled in communication with each other to form an active/passive drive device. Moreover, in the principle that the driving force and the load resistance must be balanced, during the operation process, the drive device can automatically adjust the rotational speed ratio between the active pump and the passive pump as a transmission drive device.

2. Description of the Related Art

(2) The conventional pumps can be generally classified into two major types, that is, the pump with constant suction/discharge amount and the pump with variable suction/discharge amount. The pump with variable suction/discharge amount has wider application range and thus is popularly employed in relevant industries. With respect to the structural form, the pump with variable suction/discharge amount can be further classified into two types, that is, piston-type pump with variable suction/discharge amount and rotary vane pump with variable suction/discharge amount. The piston-type pump with variable suction/discharge amount generally has a rotary swash plate with variable angle. In rotation, the swash plate sequentially pushes multiple piston-type cylinder blocks arranged substantially in parallel to each other. FIG. **1** shows a conventional rotary vane pump with variable suction/discharge amount. The rotary vane pump mainly includes a vane rotor **10** disposed in a cam ring **11** inside the pump **1**. An eccentric amount adjustment member **12** is disposed on one side of the cam ring **11** to push the cam ring **11** and adjust the eccentric amount of the eccentric amount adjustment member **12** to the vane rotor **10**. The eccentric amount is adjustable so that the fluid receiving space between the vane rotor **10** and the cam ring **11** can be modulated so as to vary the suction/discharge amount of the pump.

(3) However, the cam ring **11** is mounted in the pump **1** so that the adjustable displacement amount is limited within the fixed space of the housing of the pump. The size of the internal space of the housing directly affects and restricts the radial sizes of the pump body and all the components. As a result, when it is necessary to manufacture different products with maximal suction/discharge amount, the commonality of the components of the different pumps with different suction/discharge amounts is quite low. Therefore, it is necessary redesign numerous components of each new pump with maximal suction/discharge amount and manufacture the molds for molding the components. As a result, the manufacturing cost is greatly increased. In addition, in operation, in case the distance between the suction side and the discharge side of the pump is relatively long, then the pressure difference between the suction side and the discharge side will be excessively great. Under such circumstance, the reciprocal radial extension/retraction displacement amount of the respective vanes may be too large. This will lead to ill affection of vibration or collision noise.

SUMMARY OF THE INVENTION

(4) It is therefore a primary object of the present invention to provide a novel rotary vane pump with variable suction/discharge amount to solve the above problems existing in the conventional pump with variable suction/discharge amount. The vane chamber of the pump is extendable/retractable in an axial direction of the vane rotor to modulate the capacity of the vane chamber. Accordingly, the unit circulation suction/discharge amount of the fluid in the pump can be increased/decreased with the axial change of the space of the vane chamber. Therefore, when it is necessary to manufacture different pumps with different suction/discharge amounts, the radial specifications of the respective components are in conformity with each other so that the community in use of the components is enhanced and the manufacturing and material costs of different pumps with suction/discharge amounts are greatly lowered. Moreover, when the requirement for the maximal suction/discharge amount of the pump is increase, it is only necessary to modify the axial size of the pump and the relevant components without enlarging the radial size of the vane chamber to increase the pressure difference between the suction side and the discharge side in the vane chamber. Also, the radial extending/retracting travel of the vane will not be elongated due to the increase of the radial size of the vane chamber. Therefore, in operation, the noise made by the reciprocal extension/retraction of the vane can be effectively lowered.
(5) It is a further object of the present invention to provide a transmission drive device composed of at least two pumps with variable suction/discharge amount. The two pumps are oppositely arranged. The fluid suction passage of one of the two pumps is in communication with the fluid discharge passage of the other of the two pumps to form a closed driving loop for the active pump to drive the passive pump. During the driving operation process of the loop, when a difference value exists between the driving force of the active pump and the load resistance of the passive pump, the difference value pushes and acts on the extendable/retractable vane chamber of the vane chamber body, whereby the capacity of the vane chamber of the active pump and the capacity of the vane chamber of the passive pump are automatically extended/retracted and modulated until the driving force applied to the fluid in the active pump and the load resistance pushed by the fluid in the passive pump are balanced. Also, in the condition that the fluid suction/discharge amount per unit time of the active pump and the fluid suction/discharge amount per unit time of the passive pump are nearly equal to each other, the capacities of the vane chambers and the rotational speeds of the active pump and the passive pump are automatically adjusted to be in inverse proportion to each other so as to balance the operation. Therefore, when the driving force or the load resistance changes, the rotational speed ratio of the active pump and the passive pump is automatically adjusted according to the change of the driving force and the load resistance so as to achieve the object of smooth transmission driving.
(6) To achieve the above and other objects, the pump with variable suction/discharge amount of the present invention includes a vane chamber body and a vane rotor disposed in the vane chamber body. The vane chamber body is at least composed of a fixed wall member, a movable wall member and a movable vane chamber sleeve, which define a vane chamber. The vane rotor has an impeller disposed in the vane chamber. At least one vane is disposed on the impeller. The movable wall member and the movable vane chamber sleeve are displaceable in an axial direction of the vane rotor relative to the fixed wall member, whereby the vane chamber is extendable/retractable in the axial direction of the vane rotor to increase/decrease the capacity of the vane chamber.
(7) In the above pump with variable suction/discharge amount, the number of the vanes is less than or equal to the number of the eccentric vane chamber sections.
(8) In the above pump with variable suction/discharge amount, one single vane is disposed on the impeller and the vane chamber has at least one eccentric vane chamber section.
(9) In the above pump with variable suction/discharge amount, the vane chamber has multiple eccentric vane chamber sections and multiple vanes are disposed on the impeller in adaptation to the multiple eccentric vane chamber sections.
(10) In the above pump with variable suction/discharge amount, two rotor shaft ends of the vane

rotor are respectively disposed on two support bodies corresponding to the two rotor shaft ends. At least one of the rotor shaft ends is externally connected with a driving member for receiving power or bearing load.

(11) In the above pump with variable suction/discharge amount, the fixed wall member has a fixed wall seat sleeve and a fixed wall end face. The fixed wall end face is disposed at one end of the fixed wall seat sleeve and normal to the axis of the vane rotor, whereby the fixed wall end face can tightly attach to an end face of the impeller of the vane rotor normal to the axial direction of the vane rotor.

(12) In the above pump with variable suction/discharge amount, the fixed wall member is fitted on a base seat of the support body (at one end).

(13) The base seat has a fixed wall end boss. The fixed wall end boss is fully plugged in a fixed wall hole formed at a center of the fixed wall end face, whereby a boss end face of the fixed wall end boss and the fixed wall end face together form a fixed wall face and the fixed wall face can tightly attach to an end face of the vane rotor normal to the axial direction of the vane rotor. An eccentric rotor shaft hole is formed on the fixed wall end boss. A shaft end of the vane rotor is pivotally fitted in the eccentric rotor shaft hole.

(14) In the above pump with variable suction/discharge amount, at least two fluid suction/discharge passages are formed in the vane rotor. One end of each suction/discharge passage, which end is directed to the vane chamber, is in communication with a suction side and a discharge side of the vane of the vane rotor. One end of each suction/discharge passage, which end is distal from the suction side and the discharge side, is in communication with at least one of two rotor shaft ends of the vane rotor.

(15) In the above pump with variable suction/discharge amount, a fluid suction/discharge port member is pivotally fitted on and assembled with the rotor shaft end in communication with the suction/discharge passages, whereby the rotor shaft end can pivotally rotate in the fluid suction/discharge port member, while the fluid suction/discharge port member is disposed on a support body (at one end) and keeps stationary.

(16) In the above pump with variable suction/discharge amount, the suction/discharge passages extend to the same rotor shaft end. The suction/discharge passages respectively communicates with a central section and a non-central section of the rotor shaft end and connecting with outer side directly via a suction/discharge passage disposed on at least one of the base seat and the support body.

(17) In the above pump with variable suction/discharge amount, the movable wall member is fitted around and assembled with the vane rotor, whereby the movable wall member can slide on an outer circumference of the impeller in the axial direction of the vane rotor, the movable vane chamber sleeve being fitted around the fixed wall member and the impeller to synchronously axially slide with the movable wall member.

(18) In the above pump with variable suction/discharge amount, a retainer member is assembled between the movable wall member and the movable vane chamber sleeve so as to keep the movable wall member and the movable vane chamber sleeve attach to and assemble with each other, whereby the movable wall member and the movable vane chamber sleeve can synchronously slide in the axial direction of the vane rotor.

(19) In the above pump with variable suction/discharge amount, the movable wall member has a movable wall face. The movable wall face tightly attaches to vane chamber sleeve end face of the movable vane chamber sleeve distal from the fixed wall member. A fitting hole is formed at a center of the movable wall member, which is axially slidable to fit around the impeller. An inner wall of the fitting hole is formed with a vane receiving slot corresponding to the vane of the impeller, whereby the vane can slide into the vane receiving slot.

(20) In the above pump with variable suction/discharge amount, the vane chamber inside the movable vane chamber sleeve is defined between the movable vane chamber sleeve, the fixed wall

end face of the fixed wall member, the movable wall face of the movable wall member and the vane rotor. The impeller occupying a part of the vane chamber and the remaining space of the vane chamber forms at least one eccentric vane chamber section eccentric to the axis of the vane rotor.
(21) In the above pump with variable suction/discharge amount, the vane has a vane top edge distal from the vane rotor. The vane top edge tightly attaches to the inner wall of the vane chamber and is slidable relative to the inner wall of the vane chamber in at least one of the axial and circumferential directions of the vane rotor.
(22) In the above pump with variable suction/discharge amount, a sealing block is disposed at inter-contacting sections of the vane, the movable wall member and the movable vane chamber sleeve to avoid any gap between the inter-contacting sections of the vane, the movable wall member and the movable vane chamber sleeve, whereby the fluid in the vane chamber is prevented from leaking.
(23) In the above pump with variable suction/discharge amount, an operation fluid is output and input into the vane chamber in a closed loop, at least one of the movable wall member and the movable vane chamber sleeve being displaceable relative to the fixed wall member, whereby the capacity of the vane chamber is changeable and the output amount and input amount of the operation fluid pushed by the rotating vane rotor to pass the vane chamber per unit time are variable with the change of the capacity of the vane chamber, whereby the vane rotor can provide power transmission at different rotational speeds according to the change of the capacity of the vane chamber.
(24) In the above pump with variable suction/discharge amount, in the push transfer process of the sole operation fluid, the pressure in the vane chamber is changed, the change amount of the pressure pushing and acting between the movable wall member, the movable vane chamber sleeve, the vane rotor and the fixed wall member, whereby at least the movable wall member and the fixed wall member are displaced relative to each other.
(25) In the above pump with variable suction/discharge amount, an external forcing member applies a push force to at least one of the movable wall member and the movable vane chamber sleeve to forcedly at least make the movable wall member and the fixed wall member displace relative to each other.
(26) In the above pump with variable suction/discharge amount, a transmission drive device composed of the above pump with variable suction/discharge amount of the present invention is composed of at least two pumps with variable suction/discharge amount. The two pumps with variable suction/discharge amount are oppositely arranged. One of the two pumps with variable suction/discharge amount is set a active pump, while the other of the two pumps with variable suction/discharge amount is set a passive pump, a driving loop being formed between the active pump and the passive pump.
(27) In the above transmission drive device, at least one of a same-direction displacement connection member and a synchronous displacement connection member is drivingly connected between at least one of the movable wall member and the movable vane chamber sleeve of the active pump and the movable wall member and the movable vane chamber sleeve of the passive pump.
(28) In the above transmission drive device, a displacement resistant member is additionally arranged in at least one of the increasing direction of the capacity of the vane chamber of the active pump and the decreasing direction of the capacity of the vane chamber of the passive pump.
(29) In the above transmission drive device, each of the active pump end and the passive pump end has at least four-time vanes and a number of eccentric vane chamber sections, which number is more than or equal to the number of the vanes. The angle phase of each vane in the vane chamber corresponding to the eccentric vane chamber section is 180-degree different from the angle phase of at least another vane in the vane chamber in a complementary relationship.
(30) In the above transmission drive device, each the four-time eccentric vane chamber sections are integrally formed in one single pump with variable suction/discharge amount.

(31) In the above transmission drive device, each eccentric vane chamber sections are formed in each independent pump with variable suction/discharge amount.
(32) In the above transmission drive device, the transmission drive device has at least two active pumps and a common engagement member is engaged between the two active pumps to synchronously drive the two active pumps.
(33) In the above transmission drive device, the transmission drive device has at least two active pumps and a common engagement member is engaged around the two active pumps to synchronously drive the two active pumps.
(34) In the above transmission drive device, the transmission drive device has at least two active pumps and at least two passive pumps. At least one of the active pumps and the passive pumps is assembled and connected in an array.
(35) In the above transmission drive device, the transmission drive device has at least two active pumps and at least two passive pumps. At least one of the active pumps and the passive pumps is linearly assembled and connected.
(36) In the above transmission drive device, the transmission drive device has at least two active pumps and at least two passive pumps. At least one of the active pumps and the passive pumps is serially assembled and connected in the form of a string.
(37) In the above transmission drive device employing the pump with variable suction/discharge amount of the present invention, an operation fluid is output and input into the vane chamber in a closed loop, at least one of the movable wall member and the movable vane chamber sleeve being displaceable relative to the fixed wall member, whereby the capacity of the vane chamber is changeable and the output amount and input amount of the operation fluid pushed by the rotating vane rotor to pass the vane chamber per unit time are variable with the change of the capacity of the vane chamber, whereby the vane rotor can provide power transmission at different rotational speeds according to the change of the capacity of the vane chamber.
(38) In the above transmission drive device, in the push transfer process of the sole operation fluid, the pressure in the vane chamber is changed, the change amount of the pressure pushing and acting between the movable wall member, the movable vane chamber sleeve, the vane rotor and the fixed wall member, whereby at least the movable wall member and the fixed wall member are displaced relative to each other.
(39) In the above driving method, an external forcing member applies a push force to at least one of the movable wall member and the movable vane chamber sleeve to forcedly at least make the movable wall member and the fixed wall member displace relative to each other.
(40) The driving method employing the pump with variable suction/discharge amount of the present invention includes steps of: (1) making the transmission drive device operate and producing a difference value between the driving force of the active pump and the load resistance born by the passive pump; (2) automatically extending/retracting and modulating the capacity of the vane chamber of the active pump and the capacity of the vane chamber of the passive pump due to the push of the difference value between the driving force and load resistance; and (3) in the condition that the fluid suction/discharge amount per unit time of the active pump and the fluid suction/discharge amount per unit time of the passive pump are nearly equal to each other, making the driving force applied to the fluid in the active pump equal to the load resistance of the fluid in the passive pump so as to achieve balanced operation and automatically adjusting the capacity of the vane chamber of the active pump and the capacity of the vane chamber of the passive pump and the rotational speeds of the active pump and the passive pump to make the active pump and the passive pump operate in inverse proportion to each other, whereby when the action between the driving force and the load resistance changes, the capacity of the vane chamber of the active pump and the capacity of the vane chamber of the passive pump and the rotational speed ratio therebetween are automatically adjusted so as to achieve balanced driving between the driving force and the load resistance in operation.

(41) In the above driving method, in operation of the driving loop, the driving force of the active pump rotates the vane rotor to drive the vane to apply a push pressure to the movable vane chamber sleeve, the fixed wall face of the fixed wall member and the movable wall face of the movable wall member positioned on the discharge side of the vane in the eccentric vane chamber section of the active pump and the vane face of the vane, the movable vane chamber sleeve, the fixed wall face and the movable wall face positioned on the suction side of the vane in the eccentric vane chamber section of the passive pump. On the other hand, at the same time, after pushed, a vacuum sucking force is applied to the movable vane chamber sleeve, the fixed wall face and the movable wall face positioned on the suction side of the vane in the eccentric vane chamber section of the active pump and the vane face of the vane, the movable vane chamber sleeve, the fixed wall face and the movable wall face positioned on the discharge side of the vane in the eccentric vane chamber section of the passive pump. After the movable wall face bears the push pressure or the vacuum sucking force, the movable wall member and the movable vane chamber sleeve tightly attaching thereto are synchronously axially moved. Two sides of the vane in the passive pump are respectively double-affected by the push pressure and the vacuum sucking force in the same direction and passive, whereby the vane rotor is passive to rotate and output power to the load end of the passive pump.
(42) In the above driving method, in case the area of the movable wall face of the movable wall member on the discharge side of the vane in the eccentric vane chamber section of the active pump is larger than the area of the movable wall face of the movable wall member on the suction side of the vane in the eccentric vane chamber section of the passive pump, the movable wall member and the movable vane chamber sleeve of the active pump gradually axially displace in a direction away from the fixed wall face of the fixed wall face to enlarge the axial space of the eccentric vane chamber section. At the same time, a sucking force is applied to the suction side of the vane of the passive pump, whereby the movable wall member and the movable vane chamber sleeve of the passive pump are sucked to axially displace in a direction toward the fixed wall face. Also, in case the area of the movable wall face on the suction side of the vane in the eccentric vane chamber section of the active pump is smaller than the area of the movable wall face on the discharge side of the vane in the eccentric vane chamber section of the passive pump, the vane of the active pump sweeps to produce vacuum sucking force on the suction side. A greater vacuum sucking force is applied to the movable wall face in the passive pump with larger area, whereby the movable wall member and the movable vane chamber sleeve of the active pump displace in a direction away from the fixed wall face and the movable wall member and the movable vane chamber sleeve of the passive pump displace in a direction toward the fixed wall face. Reversely, when the sizes of the areas of the movable wall faces on the discharge side and the suction side of the vanes respectively in the eccentric vane chamber sections of the active pump and the passive pump are compared with each other to be on the contrary to the above, the movable wall member and the movable vane chamber sleeve of the active pump and the passive pump displace in a direction reverse to the above direction. The same-direction displacement connection member is connected between the active pump and the passive pump so that along with the driving of the vane of the active pump. The liquid phase fluid applies a push force to the vane face on the suction side of the vane in the passive pump to gradually push the passive pump and the load end thereof so that the driving loop of the active pump and the passive pump will gradually start to operate.
(43) In the above driving method, the active pump assembly with multiple eccentric vane chamber sections is connected with the passive pump assembly with multiple eccentric vane chamber sections. The sums of the areas of the movable wall faces respectively on two sides of the vanes in the eccentric vane chamber sections of the active pump and the passive pumps are equal to each other, whereby the driving force of the active pump assembly is balanced with the load resistance of the passive pump assembly and the sums of the areas of the movable wall faces on the discharge sides and the suction sides of the active pump assembly and the passive pump assembly are equal

to each other. Also, the angle phases of the vanes respectively positioned in the eccentric vane chamber sections is arranged in a corresponding complementary relationship, whereby during any operation process, the active pump assembly and the passive pump assembly always has a vane face for bearing the power to provide driving effect.

(44) In the above driving method, in operation of the closed driving loop of at least one of the active pump assembly and the passive pump assembly, when the driving force and the load resistance are varied, the total capacity and rotational speed of the active pump and the total capacity and rotational speed of the passive pump are automatically adjusted to make the driving force and the load resistance automatically achieve a balanced state. The rotational speed ratio of the active pump and the passive pump is automatically modulated according to the change of the driving force and the load resistance.

(45) The present invention can be best understood through the following description and accompanying drawings, wherein:

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a sectional view of a conventional pump with variable suction/discharge amount, showing the structure thereof;

(2) FIG. **2** is a perspective exploded view of a first preferred embodiment of the present invention;

(3) FIG. **3** is a perspective partially assembled view of the first preferred embodiment of the present invention according to FIG. **2**;

(4) FIG. **4** is a sectional assembled view of the first preferred embodiment of the present invention according to FIG. **2**, showing that the space of the vane chamber is relatively smaller than the space of the vane chamber of FIG. **4-1**;

(5) FIG. **4-1** is a sectional assembled view of the first preferred embodiment of the present invention according to FIG. **2**, showing that the space of the vane chamber is relatively larger than the space of the vane chamber of FIG. **4**;

(6) FIG. **5** is a sectional assembled view of the first preferred embodiment of the present invention according to FIG. **2**, showing that a forcing mechanism is used to drive the movable wall member;

(7) FIG. **6** is a sectional assembled view of the first preferred embodiment of the present invention according to FIG. **2**, showing that the suction passage and discharge passage respectively communicate with outer side via two shaft ends of the vane rotor;

(8) FIG. **7** is a sectional assembled view of the first preferred embodiment of the present invention according to FIG. **2**, showing that a active pump is assembled with a passive pump, wherein a same-direction displacement connection member is connected between at least one of the movable wall member and the movable vane chamber sleeve of the active pump and the passive pump;

(9) FIG. **7-1** is a sectional assembled view of the first preferred embodiment of the present invention according to FIG. **2**, showing that two active pumps are assembled with two passive pumps, wherein a same-direction displacement connection member is connected between at least one of the movable wall member and the movable vane chamber sleeve of the active pump and the passive pump;

(10) FIG. **7-2** is a sectional assembled view of the first preferred embodiment of the present invention according to FIG. **2**, showing that four active pumps are assembled with four passive pumps, wherein a synchronous displacement connection member is connected between at least one of the movable wall member and the movable vane chamber sleeve of the active pump and the passive pump;

(11) FIG. **7-3** is a sectional assembled view of the first preferred embodiment of the present invention according to FIG. **7**, wherein a displacement resistant member is additionally arranged in

the increasing direction of the capacity of the vane chamber of the active pump and a same-direction displacement connection member is connected between at least one of the movable wall member and the movable vane chamber sleeve of the active pump and the passive pump;
(12) FIG. **7-4** is a sectional assembled view of the first preferred embodiment of the present invention according to FIG. **7**, wherein a displacement resistant member is additionally arranged in the decreasing direction of the capacity of the vane chamber of the passive pump and a same-direction displacement connection member is connected between at least one of the movable wall member and the movable vane chamber sleeve of the active pump and the passive pump;
(13) FIG. **8** is a sectional assembled view of the first preferred embodiment of the present invention according to FIG. **2**, wherein two pumps are assembled to form a active pump end and a common engagement member is engaged between the two pumps to synchronously drive the two pumps;
(14) FIG. **8-1** is a sectional assembled view of the first preferred embodiment of the present invention according to FIG. **2**, wherein four pumps are assembled in an array to form a active pump end and a common engagement member is positioned at the center of the array and engaged with the four pumps to synchronously drive the four pumps;
(15) FIG. **8-2** is a sectional assembled view of the first preferred embodiment of the present invention according to FIG. **2**, wherein four pumps are assembled in an array to form a active pump end and a common engagement member is positioned around the array and engaged with the four pumps to synchronously drive the four pumps;
(16) FIG. **8-3** is a sectional assembled view of the first preferred embodiment of the present invention according to FIG. **2**, wherein four pumps are assembled to form a linearly arranged active pump end;
(17) FIG. **8-4** is a sectional assembled view of the first preferred embodiment of the present invention, wherein after the forms of a shaft end and the fluid suction port member and the fluid discharge port member are changed, four pumps are serially assembled to form a stringed active pump end;
(18) FIG. **9** is a perspective exploded view of a second preferred embodiment of the present invention;
(19) FIG. **10** is a perspective partially assembled view of the second preferred embodiment of the present invention according to FIG. **9**;
(20) FIG. **11** is an axially sectional assembled view of the second preferred embodiment of the present invention according to FIG. **9**;
(21) FIG. **12** is a radially sectional assembled view of the second preferred embodiment of the present invention according to FIG. **11**, which is taken along line A-A; and
(22) FIG. **13** is a sectional assembled view of the second preferred embodiment of the present invention according to FIG. **10**, wherein a active pump is assembled with a passive pump.
REFERENCE NUMBERS OF DRAWINGS
(23) **1** pump **10** vane rotor **101** active pump **102** passive pump **11** cam ring **12** eccentric amount adjustment member **2** vane chamber body **204** fixed wall face **21** fixed wall member **211** fixed wall seat sleeve **212** fixed wall end face **213** fixed wall hole **22** movable wall member **221** movable wall face **222** fitting hole **2221** vane receiving slot **23** movable vane chamber sleeve **230** vane chamber **2301**, **2303** eccentric vane chamber section **2302** vane chamber sleeve end face **3** vane rotor **30** impeller **301** end face **31** vane **311** vane top edge **33** first rotor shaft **34** second rotor shaft **341** first suction/discharge ports **342** second suction/discharge ports **343** first suction/discharge passages **344** second suction/discharge passages **345** shaft center **346** shaft non-center **35** fluid suction/discharge port member **351** first suction/discharge passage **352** second suction/discharge passage **36** transmission member **37** sealing block **4** first support body; **40** second support body; **41** base seat **410**, **4100** suction/discharge passage **411** fixed wall end boss **4110** boss end face **412** shaft hole **5** retainer member **6**, **60**, **61**, **62** common engagement member **8** external forcing member **80** same-direction displacement connection member **800** synchronous displacement connection member **9**,

90 displacement resistant member

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

(24) Please refer to FIGS. **2** to **4**. The present invention is mainly composed of a vane chamber body **2**, a vane rotor **3**, a first support body **4** and a second support body **40**. The vane chamber body **2** is at least composed of a fixed wall member **21**, a movable wall member **22** and a movable vane chamber sleeve **23**. The movable wall member **22** and the movable vane chamber sleeve **23** are movable in an axial direction of the vane rotor **3** and displaceable relative to the fixed wall member **21**. At least one vane chamber **230** is defined between the fixed wall member **21**, the movable wall member **22** and the movable vane chamber sleeve **23**. When the movable wall member **22** and the movable vane chamber sleeve **23** are moved in the axial direction of the vane rotor **3** and displaced relative to the fixed wall member **21**, the capacity of the vane chamber **230** is changed.

(25) According to the above principle, in a first embodiment of the present invention (as shown in FIGS. **2** to **5**), the fixed wall member **21** has two parts of fixed wall seat sleeve **211** and fixed wall end face **212**. The fixed wall end face **212** is disposed at one end of the fixed wall seat sleeve **211** and normal to the axis of the vane rotor **3**. A fixed wall hole **213** is formed at a center of the fixed wall end face **212**. The fixed wall seat sleeve **211** of the fixed wall member **21** is capped on a base seat **41** having a fixed wall end boss **411**. The fixed wall end boss **411** is tightly fully plugged in the fixed wall hole **213**, whereby a boss end face **4110** of the fixed wall end boss **411** and the fixed wall end face **212** together form a fixed wall face **204**. In a preferred structural form, the base seat **41** can be detachably disposed on the first support body **4** or integrally securely formed on the first support body **4**.

(26) The vane rotor **3** has at least one impeller **30** and at least one vane **31** assembled with the impeller **30**. The vane **31** is radially slidable and extendable/retractable. The impeller **30** has an end face **301** normal to the axis vane rotor **3**. The end face **301** can tightly attach to the fixed wall face **204**. The vane rotor **3** has a first rotor shaft **33**, which can be pivotally fitted in an eccentric rotor shaft hole **412** formed on the base seat **41**. The first rotor shaft **33** is further passed through the first support body **4** to externally connect with a transmission member **36** for receiving power or bearing a load. The vane rotor **3** further has a second rotor shaft **34**, in which a first suction/discharge port **341** and a second suction/discharge port **342** are formed. A first suction/discharge passage **343** and a second suction/discharge passage **344** are formed in the vane rotor **3** respectively in communication with the first and second suction/discharge ports **341**, **342**. The first and second suction/discharge passages **343**, **344** respectively extend to further communicate with a suction side and a discharge side on two sides of the vane **31** into communication with the vane chamber **230**. The second rotor shaft **34** can be directly pivotally disposed on the second support body **40**. Alternatively, as shown in FIGS. **2** to **5**, a fluid suction/discharge port member **35** can be first fitted on the second rotor shaft **34** and then the fluid suction/discharge port member **35** is disposed on the second support body **40**. The fluid suction/discharge port member **35** has a first suction/discharge passage **351** and a second suction/discharge passage **352**. The second rotor shaft **34** is pivotally fitted in the fluid suction/discharge port member **35** and rotated relative to the fluid suction/discharge port member **35**. Therefore, with the fluid suction/discharge port member **35** serving as a fluid connection interface (as shown in FIGS. **4** and **5**), the first and second suction/discharge ports **341**, **342** of the second rotor shaft **34** can correspondingly communicate with the first and second suction/discharge passages **351**, **352** of the fluid suction/discharge port member **35**, whereby the first and second suction/discharge ports **341**, **342** and the internal fluid passages of the second rotor shaft **34** can be converted from an original rotating state into a stationary state. Accordingly, in continuous operation of the vane rotor **3**, the first and second suction/discharge ports **341**, **342** and the internal fluid passages of the second rotor shaft **34** can keep in connection with an external fluid input source and an external fluid output source. The first and second suction/discharge passages **343**, **344** in the vane rotor **3** can have various forms in

addition to the above form. For example, as shown in FIG. **6**, the first and second suction/discharge passages **343**, **344** can communicate with outer side via the first and second rotor shafts **33**, **34** of the vane rotor **3**. Alternatively, as shown in FIG. **11**, the first and second suction/discharge passages **343**, **344** can respectively communicate with a shaft center **345** and shaft non-center **346** of the second rotor shaft **34** and then connect with the outer side directly via a suction/discharge passage **410** and a suction/discharge passage **4100** disposed on the base seat **41** and/or the first support body **4**.

(27) The movable wall member **22** is fitted around the vane rotor **3** and is axially slidable to fit around the impeller **30**. The movable wall member **22** has a movable wall face **221**. The movable wall face **221** is tightly attached to a vane chamber sleeve end face **2302** of the movable vane chamber sleeve **23**, which faces the movable wall member **22**. A fitting hole **222** is formed at a center of the movable wall member **22**, which is axially slidable to fit around the impeller **30**. An inner wall of the fitting hole **222** is formed with a vane receiving slot **2221** corresponding to the vane **31**. The vane **31** can slide into the vane receiving slot **2221**, whereby when the movable wall member **22** relatively axially approaches the fixed wall member **21**, more part of the vane **31** can slide into the vane receiving slot **2221**. The vane chamber **230** is defined in the movable vane chamber sleeve **23**. The vane chamber **230** can axially slide to fit around the fixed wall member **21** and the impeller **30**. The vane chamber **230** is defined between the movable vane chamber sleeve **23**, the fixed wall end face **212**, the movable wall face **221** and the vane rotor **3**. The impeller **30** occupies a part of the vane chamber **230**. The remaining space of the vane chamber **230** forms at least one eccentric vane chamber section **2301** eccentric to the axis of the vane rotor **3**. The vane **31** has a vane top edge **311** distal from the vane rotor **3**. The vane top edge **311** tightly attaches to the inner wall of the vane chamber **230** and is axially and/or circumferentially slidable relative to the inner wall of the vane chamber **230**. In addition, proper sealing and leakproof members can be disposed between the contacting sections of the vane **31** and the inner wall of the vane chamber **230** and between the tightly attaching or relatively displacing sections of the fixed wall member **21**, the movable wall member **22**, the movable vane chamber sleeve **23** and the vane rotor **3** so as to prevent the fluid in the operating vane chamber **230** from leaking through the aforesaid sections. Especially, at the inter-contacting sections of the vane top edge **311** of the vane **31**, the movable wall member **22** and the movable vane chamber sleeve **23**, the curve of the configuration of the vane top edge **311**, the cross-sectional curve of the vane receiving slot **2221** of the movable wall member **22**, into which the vane top edge **311** can slide and the curve of the inner wall of the vane chamber **230** of the movable vane chamber sleeve **23** in contact with the vane top edge **311** are different from each other. Therefore, minor gaps exist between the inter-contacting sections of the vane top edge **311** of the vane **31**, the movable wall member **22** and the movable vane chamber sleeve **23**. As a result, in operation, the vane chamber **230** cannot be fully closed. In order to solve this problem, a sealing block **37** is disposed on the vane top edge **311**, which can tightly attach to the vane top edge **311** to synchronously slide with the vane **31**. The sealing block **37** is further restricted in the intersection path of the vane receiving slot **2221** of the movable wall member **22** and the outer edge of the inner wall of the vane chamber **230** of the movable vane chamber sleeve **23**. Accordingly, in operation, the sealing block **37** always seals the inter-contacting sections of the vane top edge **311**, the vane receiving slot **2221** and the outer edge of the inner wall of the vane chamber **230** and blocks the gaps to achieve good sealing and leakproof effect. A retainer member **5** can be assembled between the movable wall member **22** and the movable vane chamber sleeve **23** so as to keep the movable wall member **22** and the movable vane chamber sleeve **23** attach to and assemble with each other, whereby the movable wall member **22** and the movable vane chamber sleeve **23** can synchronously axially slide. (The retainer member **5** can have various structural forms and will not be redundantly described hereinafter).

(28) According to the above assembled structure, in operation, when the vane rotor **3** drives the vane **31** to sweep within the eccentric vane chamber section **2301**, the fluid on the forward side of

the sweeping direction of the vane **31** is compressed and discharged as a discharge side. The fluid positioned on the other side of the vane **31** is sucked in as a suction side. In addition, the eccentric vane chamber section **2301** is eccentric to the axis of the vane rotor **3** so that the area of the fixed wall face **204** per unit angle, which the vane **31** sweeps over in the eccentric vane chamber section **2301**, will continuously change along with the rotation of the vane rotor **3**. This phenomenon is equivalent to that the intersection area of the movable wall face **221** on two sides of the vane **31** and the interior of the eccentric vane chamber section **2301** and the suction/discharge amount of the fluid on two sides of the vane **31** will both change along with the change of the sweeping position of the vane **31**. Also, the space of the eccentric vane chamber section **2301**, which is occupied by the vane **31**, is relatively changed. This leads to some difference between the fluid amount discharged from the discharge side of the vane **31** and the fluid amount sucked into the suction side of the vane **31**. Moreover, under the forced push of an external forcing member **8** (as shown in FIG. **5**) or under the action of the differences between the flow amount of the operation fluid and the pressure, the movable wall member **22** in association with the movable vane chamber sleeve **23** is fitted on the vane rotor **3** and the fixed wall seat sleeve **211** to relatively axially displace. When the movable wall face **221** gradually axially gets close to the fixed wall face **204**, the available suction/discharge capacity of the eccentric vane chamber section **2301** is relatively gradually reduced. Reversely, when the movable wall face **221** gradually axially moves away from the fixed wall face **204**, the available suction/discharge capacity of the eccentric vane chamber section **2301** is gradually increased. Accordingly, a pump with variable suction/discharge amount, which is axially extendable/retractable to change the suction/discharge amount of the vane chamber **230**, is formed.
(29) Accordingly, the above pump with variable suction/discharge amount can be applied to and assembled with a closed loop. The closed loop outputs and inputs an operation fluid to the vane chamber **230** and the external forcing member **8** forcedly pushes the pump to transfer the operation fluid. In the transfer process of the operation fluid, the pressure in the vane chamber **230** is changed. The change amount of the pressure acts between at least one of the movable wall member **22** and the movable vane chamber sleeve **23** and the fixed wall member **21**, whereby the movable wall member **22** and/or the movable vane chamber sleeve **23** and the fixed wall member **21** displace relative to each other so as to change capacity of the vane chamber **230**. Accordingly, the output amount and input amount of the operation fluid pushed by the rotating vane rotor **3** to pass the vane chamber **230** per unit time are variable with the change of the capacity of the vane chamber **230**, whereby the vane rotor **3** can provide power transmission at different rotational speeds according to the change of the capacity of the vane chamber **230**.
(30) As shown in FIG. **7**, two pumps with variable suction/discharge amount of the present invention are oppositely arranged in communication with each other. The suction port and discharge port of the first suction/discharge passage **351** and second suction/discharge passage **352** of the two oppositely arranged pumps are in communication with each other. Accordingly, in case the pump of the two oppositely arranged pumps on the left side of the drawing is set a active pump **101**, while the pump on the right side is set a passive pump **102** and the discharge passage of the active pump **101** is in communication with the suction passage of the passive pump **102**, the fluid discharged from the discharge passage of the active pump **101** can enter the suction passage and the suction side of the vane **31** of the passive pump **102**. Reversely, in case the discharge passage of the passive pump **102** is in communication with the suction passage of the active pump **101**, the fluid on the discharge side of the vane **31** of the passive pump **102** is discharged from the discharge passage and then flows back to the suction passage and the suction side of the vane **31** of the active pump **101**, whereby the vane chambers and the entire suction and discharge passages of the active pump **101** and the passive pump **102** form a close loop for the active pump **101** to drive the passive pump **102**. In addition, a same-direction displacement connection member **80** is connected between at least one of the movable wall member **22** and the movable vane chamber sleeve **23** of the active

pump **101** and the passive pump **102**, whereby the movable wall member **22** and the movable vane chamber sleeve **23** of the active pump **101** and the passive pump **102** can move together in the same axial direction. In operation of the closed loop of the active pump **101**, in case the employed fluid is a liquid phase fluid and the total volume of the liquid is constant, then the liquid phase fluid on the discharge side in the eccentric vane chamber section **2301** of the active pump **101** will be pushed by the vane **31** of the rotating vane rotor **3** to the suction side of the passive pump **102**. Relatively, the liquid phase fluid on the discharge side in the eccentric vane chamber section **2301** of the passive pump **102** will be pushed by the vane **31** of the rotating vane rotor **3** to the suction side of the active pump **101**. Accordingly, a complete liquid phase fluid driving loop of the active pump and the passive pump is formed. In operation of the driving loop, the driving force of the active pump **101** rotates the vane rotor **3** to drive the vane **31** to apply a push pressure to the movable vane chamber sleeve **23**, the fixed wall face **204** and the movable wall face **221** positioned on the discharge side of the vane **31** in the eccentric vane chamber section **2301** of the active pump **101** and the vane face of the vane **31**, the movable vane chamber sleeve **23**, the fixed wall face **204** and the movable wall face **221** positioned on the suction side of the vane **31** in the eccentric vane chamber section **2301** of the passive pump **102**. On the other hand, after pushed, a vacuum sucking force is applied to the movable vane chamber sleeve **23**, the fixed wall face **204** and the movable wall face **221** positioned on the suction side of the vane **31** in the eccentric vane chamber section **2301** of the active pump **101** and the vane face of the vane **31**, the movable vane chamber sleeve **23**, the fixed wall face **204** and the movable wall face **221** positioned on the discharge side of the vane **31** in the eccentric vane chamber section **2301** of the passive pump **102**. The direction of the push pressure or vacuum sucking force applied to the movable vane chamber sleeve **23** is normal to the axial moving direction of the movable vane chamber sleeve **23** so that the push pressure or vacuum sucking force cannot directly make the movable vane chamber sleeve **23** displace. The fixed wall face **204** is fixed and unmovable. Therefore, during the driving process, only the movable wall face **221** will bear the push pressure or vacuum sucking force to make the movable wall member **22** axially move. At the same time, the movable vane chamber sleeve **23** is passive to tightly attach to the movable wall member **22** and synchronously axially move. Two sides of the vane **31** in the passive pump **102** are respectively double-affected by the push pressure and the vacuum sucking force in the same direction, whereby the vane **31** is passive to drive and rotate the vane rotor **3** so as to output power to the load end of the passive pump **102**. At the beginning of the driving process, the passive pump **102** is situated in a stationary state. The vane rotor **3** of the active pump **101** starts to be rotated under the driving force, whereby the liquid phase fluid on the discharge side of the vane **31** starts to be pushed and compressed. At this time, in case the area of the movable wall face **221** on the discharge side of the vane **31** in the eccentric vane chamber section **2301** of the active pump **101** is larger than the area of the movable wall face **221** on the suction side of the vane **31** in the eccentric vane chamber section **2301** of the passive pump **102**, due to that the larger the forced area is, the greater the push pressure applied to the forced area is and due to that a load force is applied to the vane **31** of the passive pump **102**, then the movable wall member **22** and the movable vane chamber sleeve **23** of the active pump **101** will gradually axially displace in a direction away from the fixed wall face **204** to enlarge the axial space of the eccentric vane chamber section **2301**. At the same time, a sucking force is applied to the suction side of the vane **31** of the passive pump **102**, whereby the movable wall member **22** and the movable vane chamber sleeve **23** of the passive pump **102** are sucked to axially displace in a direction toward the fixed wall face **204**. At this time, the area of the movable wall face **221** on the suction side of the vane **31** in the eccentric vane chamber section **2301** of the active pump **101** is smaller than the area of the movable wall face **221** on the discharge side of the vane **31** in the eccentric vane chamber section **2301** of the passive pump **102**. Therefore, after the vane **31** of the active pump **101** sweeps, the vacuum sucking force applied to the suction side of the vane **31** provides greater sucking driving force for the movable wall face **221** in the passive pump **102** with larger area. As a result, the movable wall member **22** and the movable vane chamber sleeve **23** of the active pump **101** will displace in a direction away from the fixed wall face **204**. The movable wall member **22** and the movable vane chamber sleeve **23** of the passive pump **102** will displace in a direction toward the fixed wall face **204**. Similarly, when the sizes of the areas of the movable wall faces **221** on the discharge side and the suction side of the vane **31** are compared with each other to be on the contrary to the above, the movable wall member **22** and the movable vane chamber sleeve **23** of the active pump **101** and the passive pump **102** will displace in a direction reverse to the above direction. During the operation process of the closed loop, the movable wall member **22** and the movable vane chamber sleeve **23** will continuously reciprocally axially displace as aforesaid until the liquid phase fluid originally on the suction side of the vane **31** of the active pump **101** and the liquid phase fluid originally in the passage of the discharge side of the vane **31** of the passive pump **102** are passive and circulated and switched to be respectively on the discharge side of the vane **31** of the active pump **101** and in the passage of the suction side of the vane **31** of the passive pump **102**. In addition, after switched, the volume of the liquid phase fluid in the passage has become larger than the sum of the allowable modulated maximal capacity on the discharge side of the vane **31** of the active pump **101** and the suction side of the vane **31** of the passive pump **102** by means of axial displacement. The same-direction displacement connection member **80** is connected between the active pump **101** and the passive pump **102** and the liquid is uncompressible so that along with the driving of the vane **31** of the active pump **101**, the vane face on the suction side of the vane **31** in the passive pump **102** will entirely bear the push force of the liquid phase fluid to gradually push and the passive pump **102** and the load end thereof. Therefore, the active/passive pump closed loop will gradually start to operate.

(31) Therefore, in application of the transmission drive device composed of the above components, in case the closed loop outputs and inputs an operation fluid to the respective vane chambers **230** of the active pump **101** and the passive pump **102**, by means of the forced push of the external forcing member **8** or the change amount of the pressure applied to the interior of the vane chamber **230** by the operation fluid during the push and transfer process, the push acts between at least one of the movable wall member **22** and the movable vane chamber sleeve **23** and the fixed wall member **21**, whereby the movable wall member **22** and/or the movable vane chamber sleeve **23** and the fixed wall member **21** displace relative to each other so as to change the capacity of the vane chamber **230**. Accordingly, the active pump **101** and the passive pump **102** can make the rotational speeds of the corresponding vane rotors **3** in inverse proportion to each other respectively according to the change of the capacity of the corresponding vane chambers **230** to provide power transmission.

(32) During the operation process of the active/passive pump loop, the movable wall members **22** and the movable vane chamber sleeves **23** of the active pump **101** and the passive pump **102** will continuously reciprocally axially displace. Therefore, the driving force applied to the vane **31** of the passive pump **102** by the active pump **101** will be interrupted. As a result, the rotation of the passive pump **102** will be undulated. Moreover, in the above embodiment, each of the active pump and the passive pump has one single vane chamber and one single vane. In case at the beginning of actuation of the passive pump, the vane of the passive pump is positioned in a position where the vane is right fully inlaid in the vane rotor, there is no vane face in the passive pump to bear the driving force. Under such circumstance, the active pump is situated in an invalid idling state and cannot apply any driving force to the passive pump. As a result, the entire loop will idle. In order to avoid the above condition of undulated operation or idling of the loop, as shown in FIG. **7-1**, two active pumps **101** (or a active pump **101** with two eccentric vane chamber sections **2301**) can be coupled with two passive pumps **102** (or a passive pump **102** with two eccentric vane chamber sections **2301**). Alternatively, as shown in FIG. **7-2**, four active pumps **101** (or a active pump **101** with four eccentric vane chamber sections **2301**) can be coupled with four passive pumps **102** (or a passive pump **102** with four eccentric vane chamber sections **2301**). After more active pumps and passive pumps are assembled with each other, the sums of the areas of the movable wall faces **221**

on two sides of the vane **31** in the eccentric vane chamber sections **2301** are approximately or nearly equal to each other. Accordingly, the driving force of the assembly of multiple active pumps is temporarily balanced with the load resistance of the assembly of multiple passive pumps. Under such circumstance, the sums of the areas of the movable wall faces **221** on the discharge side and the suction side of the assembly of the active pumps and the passive pumps are approximately equal to each other. This can effectively improve the above condition of undulated operation. Also, due to that the multiple pumps are assembled, the angle phases of the respective vanes **31** positioned in the eccentric vane chamber sections **2301** can be arranged in a complementary relationship. Therefore, during any operation process, the assembly of the active pumps and the passive pumps always has a vane face for bearing the power without invalidate idling phenomenon of the loop. Therefore, the entire driving process can be smoother and more stable.

(33) According to the above active/passive pump driving loop, especially the structural form composed of four active pumps **101** and four passive pumps **102** coupled therewith as shown in FIG. **7-2**, the angle phase of each vane **31** positioned in the vane chamber **230** has another symmetrical vane **31** with an angle phase 180-degree different from the vane **31** as a complementary vane. Therefore, in the assembly of the four active pumps **101** and the four passive pumps **102**, the sum of the areas of the movable wall faces **221** corresponding to the discharge side of the vane **31** in the eccentric vane chamber sections **2301** is nearly equal to the sum of the areas of the movable wall faces **221** corresponding to the suction side of the vane **31** in the eccentric vane chamber sections **2301**. This is equivalent to that the discharge amount of the liquid phase fluid in the assembly of the four active pumps **101** and the four passive pumps **102** is nearly equal to the suction amount of the liquid phase fluid in the assembly of the four active pumps **101** and the four passive pumps **102**. Accordingly, the entire loop can continuously stably operate. In the case that the driving force of the four active pumps **101** is unchanged, while the load of the four passive pumps **102** is increased, the sweeping speed of the vanes **31** of the four passive pumps **102** will be reduced. Under such circumstance, the liquid phase fluid will accumulate on the suction sides of the vanes **31** of the four passive pumps **102** to apply a capacity-enlarging push force to the movable wall faces **221**. In addition, the amount of the liquid phase fluid flowing from the discharge sides of the vanes **31** of the four passive pumps **102** back to the suction sides of the vanes **31** of the four active pumps **101** is reduced to apply a vacuum sucking force to the movable wall faces **221**. The sum of the areas of the movable wall faces **221** on the discharge side of the vane **31** in the eccentric vane chamber sections **2301** is nearly equal to the sum of the areas of the movable wall faces **221** on the suction side of the vane **31** in the eccentric vane chamber sections **2301** so that the total force applied to the movable wall faces **221** of the four active pumps **101** is nearly equal to the total force applied to the movable wall faces **221** of the four passive pumps **102**. Under the action of the capacity-enlarging push force of the four passive pumps **102** and the vacuum sucking force of the four active pumps **101**, the movable wall members **22** and the movable vane chamber sleeves **23** of the four active pumps **101** displace in a direction toward the fixed wall faces **204** to minify the total capacity of the four active pumps **101**. At the same time, the movable wall members **22** and the movable vane chamber sleeves **23** of the four passive pumps **102** displace in a direction away from the fixed wall faces **204** to enlarge the total capacity of the four passive pumps **102**. Therefore, the four active pumps **101** must circularly input the power many times so as to drive the four passive pumps **102** to circularly output the power one time. This is similar to a downshift driving effect in power transmission. Reversely, in the case that the driving force of the four active pumps **101** is unchanged, while the load of the four passive pumps **102** is reduced, all the above operation conditions are totally reversed. That is, the four active pumps **101** only need to circularly input the power one time for driving the four passive pumps **102** to circularly output the power many times. This is similar to an upshift driving effect in power transmission. It can be known from the aforesaid that in the operation of the closed driving loop composed of the four active pumps **101** and the four passive pumps **102**, when the driving force and the load resistance change, the

respective total capacities of the four active and passive pumps can be automatically adjusted so that the driving force and the load resistance can be automatically balanced with each other. Therefore, the drive device can smoothly automatically modulate the transmission according to the change of the driving force and the load resistance.

(34) As shown in FIGS. **7**, **7-1**, **7-2**, **7-3** and **7-4**, in the condition that the suction/discharge amount per unit time of the active pump **101** and the suction/discharge amount per unit time of the passive pump **102** are nearly equal to each other, a same-direction displacement connection member **80** or a synchronous displacement connection member **800** is drivingly connected between the movable wall member **22** or the movable vane chamber sleeve **23** of the active pump **101** and the passive pump **102**. An external force is applied to the same-direction displacement connection member **80** or the synchronous displacement connection member **800** to push the same so as to force the movable wall member **22** or the movable vane chamber sleeve **23** of the active pump **101** and the passive pump **102** to respectively same-direction or synchronously reversely displace away from or toward the corresponding fixed wall faces **204**. Accordingly, it can be ensured that the increase amount or the decrease amount of the capacity of the vane chamber of the active pump **101** is nearly equal to or right equal to the decrease amount or the increase amount of the capacity of the vane chamber of the passive pump **102**. In addition, a displacement resistant member **9** and a displacement resistant member **90** (such as a spring) can be additionally arranged in the increasing direction of the capacity of the vane chamber of the active pump **101** of FIG. **7-3** and the decreasing direction of the capacity of the vane chamber of the passive pump **102** of FIG. **7-4**. Accordingly, the displacement resistant member **9** and the displacement resistant member **90** can provide an internal preload resistance against the rotational speed ratio automatic regulation effect achieved between the active pumps **101** and the passive pumps **102**. Under such circumstance, the actually required input driving force needs to be slightly greater than the actually externally added load resistance. This preset balancing condition provides a forced downshift effect as a transmission mechanism.

(35) FIG. **8** shows an integrated structure of a drive device composed of two pumps connected with each other as an assembly unit. A common engagement member **6** is engaged between the two pumps to synchronously drive the two pumps. FIG. **8-1** shows an integrated structure of a drive device composed of four pumps as an assembly unit. A common engagement member **60** is engaged between the four pumps to synchronously drive the four pumps. According to the phase difference between the positions of the vanes **31** of the respective pumps in the drawings, it can be found that the suction/discharge timing between the respective pumps are just complementary to the increase/decrease of the suction/discharge amounts. Therefore, the suction/discharge amounts are equal to each other at every time point and the state is stabilized. In operation, this avoids the undulated unstable phenomenon during the driving process due to the difference between the fluid suction/discharge amounts. In addition, FIG. **8-2** shows a drive device composed of four pumps arranged in an array as an assembly unit according to FIG. **8-1**. FIG. **8-2** is simply different from FIG. **8-1** in that a common engagement member **61** is positioned around the respective pumps and engaged with the pumps to drive the pumps. This achieves a similar synchronously driving effect. Moreover, FIG. **8-3** shows a linearly arranged driving mode. A common engagement member **62** is engaged between each two adjacent pumps to linearly connect the respective pumps. FIG. **8-4** shows a stringed driving mode. The respective pumps are coaxially or nearly coaxially serially connected.

(36) Please further refer to FIGS. **9** to **12**, which show a second embodiment of the present invention. The second embodiment also mainly includes a fixed wall member **21**, a movable wall member **22** and a movable vane chamber sleeve **23** defining a vane chamber **230** having variable capacity with multiple eccentric vane chamber sections **2303**. A vane rotor **3** with multiple vanes **31** is arranged in the vane chamber **230**. The number and configuration of the vanes **31** correspond to the number and configuration of the eccentric vane chamber sections **2303**. Accordingly, a pump

with variable suction/discharge amount, which can provide many times of suction/discharge operations in one single operation cycle is achieved. In principle, the number of the vanes **31** should be less than or equal to the number of the eccentric vane chamber sections **2303** so as to prevent the suction passage and the discharge passage appear in the same eccentric vane chamber section **2303** at the same time and communicate with each other to deteriorate the driving performance of the pump.

(37) The second embodiment is most obviously different from the first embodiment in that the movable vane chamber sleeve **23** of the second embodiment can only axially displace relative to the vane rotor **3**, while failing to synchronously rotate with the vane rotor **3**. The suction/discharge passages **343**, **344** of the second embodiment can be disposed on the suction side and the discharge side of the vane **31** of the impeller **30** of the vane rotor **3** as in the first embodiment. FIG. **13** shows that the multi-vane pump with variable suction/discharge amount shown in FIGS. **9** to **12** is assembled in accordance with the assembling mode of FIG. **7**, that is, the discharge passage of the active pump **101** is in communication with the suction passage of the passive pump **102**, while the suction passage of the active pump **101** is in communication with the discharge passage of the passive pump **102**. Accordingly, as the pump of the first embodiment, a closed loop is formed between the active pump **101** and the passive pump **102** for the active pump **101** to drive the passive pump **102**. By means of the vane chamber **230** with multiple eccentric vane chamber sections **2303** and variable capacity, in operation of the closed loop, when the force difference between the driving force of the active pump **101** and the load resistance of the passive pump **102** changes, under the action of the force difference, the capacities of the eccentric vane chamber sections **2303** can be automatically modulated to a temporary balanced state after the force difference disappears. At this time, the rotational speed between the active pump **101** and the passive pump **102** is in inverse proportion to the capacity of the eccentric vane chamber sections **2303** after automatically modulated, whereby the closed loop assembly between the active pump **101** and the passive pump **102** becomes a transmission drive device capable of automatically modulating rotational speed ratio. In addition, in the second embodiment, the assembly of the active pump and the passive pump with multiple vanes **31** and multiple eccentric vane chamber sections **2303** can provide a driving force as the assembly of the multiple active pumps and the multiple passive pumps each having one single vane and one single eccentric vane chamber section as shown in FIGS. **7-1** and **7-2**. Therefore, the second embodiment can provide stable driving effect and obviously has very high utility and value in industries.

(38) According to the above design of the pump with variable suction/discharge amount of the present invention, in the condition that the original radial size is not increased, the pump with variable suction/discharge amount can truly effectively achieve the modulation function for the suction/discharge amount. The pump with variable suction/discharge amount of the present invention not only can effectively improve the shortcomings of the conventional pumps with variable suction/discharge amount, but also can be assembled to form a drive device capable of automatically modulating the rotational speed ratio between the pumps. The pump with variable suction/discharge amount of the present invention is indeed inventive and has high practical value.

(39) The above embodiments are only used to illustrate the present invention, not intended to limit the scope thereof. Many modifications of the above embodiments can be made without departing from the spirit of the present invention.

Claims

1. A pump with variable suction/discharge amount, which is characterized in including a vane chamber body and a vane rotor; the vane chamber body internally having a vane chamber, and the vane chamber having a capacity space defined between a fixed wall member, a movable wall member and a movable vane chamber sleeve in the vane chamber body, the vane chamber being

partitioned by an impeller of the vane rotor in the vane chamber into a plurality of eccentric vane chamber sections and a plurality of vanes being disposed on the impeller, and the number of the eccentric vane chamber sections being more than or equal to the number of the vanes; in the vane chamber, at any timing when the vane rotor continuously operates in the same direction relative to the vane chamber body, one of two lateral sides of each vane adjacent to another vane unchangeably being a suction side while another one of the two lateral sides of each vane adjacent to another vane unchangeably being a discharge side, and the suction side and the discharge side respectively having suction/discharge passages in communication with outer side of the pump; the fixed wall member being positioned in a fixed position in the vane chamber body; each vane having a corresponding symmetrical vane of the plurality of vanes, the angle of which is 180-degree different from the angle of the vane, whereby at any moment of the operation process; when the vane extends out of the impeller of the vane rotor, the 180-degree symmetrical vane is retracted into the impeller of the vane rotor, and when the vane is retracted into the impeller of the vane rotor, the 180-degree symmetrical vane extends out of the impeller of the vane rotor, so that the vane and the 180-degree symmetrical vane are complementary to each other; the movable wall member and the movable vane chamber sleeve being displaceable in an axial direction of the vane rotor relative to the fixed wall member to increase or decrease and change the capacity space of the vane chamber, so as to form the capacity space with variable suction/discharge amount.
2. The pump with variable suction/discharge amount as claimed in claim 1, characterized in that at least two suction/discharge passages openings are disposed on the impeller of the vane rotor between any two adjacent vanes, one of the suction/discharge passage openings being in communication with the suction side, while the other of the suction/discharge passage openings being in communication with the discharge side, and both suction/discharge passage openings being in communication with the outer side of the vane chamber.
3. The pump with variable suction/discharge amount as claimed in claim 1, characterized in that a sealing block is disposed at inter-contacting sections of the vane, the movable wall member and the movable vane chamber sleeve.
4. The pump with variable suction/discharge amount as claimed in claim 1, characterized in that the fixed wall member has a fixed wall end face, the fixed wall end face being disposed at one end of the fixed wall member, the fixed wall member being capped on a base seat of a support body, said vane rotor being formed with a rotor shaft having a first end and a second end, wherein one end of said first and second ends of the rotor shaft pass through the fixed wall member, at least one end of the first and second ends of said rotor shaft being pivotally supported on the support body, at least one end of the first and second ends of said rotor shaft outward outputting power or bearing power, the fixed wall end face being tightly attachable to an end face of the impeller of the vane rotor, the movable vane chamber sleeve being fitted on the fixed wall member around the vane rotor, the movable wall member being formed with vane receiving slots, the number of the vane receiving slots being equal to the number of the vanes, a fitting hole being formed at a center of the movable wall member, the fitting hole of the movable wall member being fitted on the impeller of the vane rotor, whereby the vanes on the impeller can slide within the vane receiving slots of the movable wall member, the movable wall member and the movable vane chamber sleeve keeping tightly attaching to each other, whereby the movable wall member and the movable vane chamber sleeve can synchronously move in the axial direction of the vane rotor to change the capacity of the vane chamber.
5. A drive device with variable suction/discharge amount composed of pumps with variable suction/discharge amount as claimed in claim 1, comprising: an active pump and a passive pump with variable suction/discharge amount oppositely coupled to one another, characterized in that said active and passive pumps with variable suction/discharge amount are connected and assembled to form the drive device with variable suction/discharge amount, wherein, during the operation process of the active pump with variable suction/discharge amount and the passive pump with

variable suction/discharge amount, the sum of the capacities of the suction sides in all the eccentric vane chamber sections being equal to the sum of the capacities of the discharge sides in all the eccentric vane chamber sections.

6. A drive device with variable suction/discharge amount composed of the pumps with variable suction/discharge amount as claimed in claim 1, characterized in that at least one of the pumps with variable suction/discharge amount are connected and assembled to form the drive device with variable suction/discharge amount, and the drive device with variable suction/discharge amount internally having the plurality of vanes, the number of which being a multiple of four.

7. A drive device with variable suction/discharge amount composed of the pumps with variable suction/discharge amount as claimed in claim 1, characterized in that at least one of the pumps with variable suction/discharge amount are connected and assembled to form an active drive device with variable suction/discharge amount and at least one pumps with variable suction/discharge amount are connected and assembled to form a passive drive device with variable suction/discharge amount, and the active drive device with variable suction/discharge amount being further connected and assembled with the passive drive device with variable suction/discharge amount to form an active/passive closed loop variable speed drive device.

8. A drive device with variable suction/discharge amount composed of the pumps with variable suction/discharge amount as claimed in claim 4, characterized in that at least one of the pumps with variable suction/discharge amount are connected and assembled to form an active drive device with variable suction/discharge amount and at least one of the pumps with variable suction/discharge amount are connected and assembled to form a passive drive device with variable suction/discharge amount, the active drive device with variable suction/discharge amount being further connected and assembled with the passive drive device with variable suction/discharge amount to form an active/passive closed loop variable speed drive device.

9. The drive device as claimed in claim 7, characterized in that a displacement resistance member is additionally disposed in at least one of enlarging directions of the space of the vane chamber of the active drive device with variable suction/discharge amount and a decreasing direction of the space of the vane chamber of the passive drive device with variable suction/discharge amount.

10. A driving method employing the pumps with variable suction/discharge amount, characterized in the following steps: in a closed loop, assembling at least one pumps with variable suction/discharge amount to form at least one drive devices with variable suction/discharge amount; the pumps with variable suction/discharge amount respectively including a vane chamber body and a vane rotor, the vane chamber body internally having a vane chamber, which has a capacity space defined between a fixed wall member, a movable wall member and a movable vane chamber sleeve in the vane chamber body the vane chamber being partitioned by an impeller of the vane rotor in the vane chamber into a plurality of eccentric vane chamber sections, a plurality of vanes being disposed on the impeller, and the number of the eccentric vane chamber sections being more than or equal to the number of the vanes; in the vane chamber, at any timing when the vane rotor continuously operates in the same direction relative to the vane chamber body, one of two lateral sides of each vanes adjacent to another vane unchangeably being a suction side while another one of the two lateral sides of each vane adjacent to another vane unchangeably being a discharge side, and the suction side and the discharge side respectively having communicable suction/discharge passages in communication with outer side of the pump; the fixed wall member being positioned in a fixed position in the vane chamber body; and the movable wall member and the movable vane chamber sleeve are displaceable in an axial direction of the vane rotor relative to the fixed wall member to increase or decrease and change the capacity space of the vane chamber, so as to form a pump capacity space with variable suction/discharge amount, inputting a fluid into the vane chamber of each drive device with variable suction/discharge amount, the input fluid pushing against one side of one of the plurality of vanes in the vane chamber to drive the vane rotor to rotate, at the same time, the fluid in the vane chamber at the other side of the vane being pushed

by the vane out of the vane chamber to form a driving loop; synchronously displacing the movable wall member and the movable vane chamber sleeve of the pump in each drive device with variable suction/discharge amount in the axial direction of the vane rotor relative to the fixed wall member so as to change the capacity space of each vane chamber of the drive device with variable suction/discharge amount, such that amounts of the fluid being pushed out of and sucked into the vane chamber change each time the vane rotor in the vane chamber rotates by one circle; inputting and outputting the same amount of fluid per unit time into and from each drive device with variable suction/discharge amount, such that the rotational speed of the vane rotor decreases when the capacity space of the vane chamber is increased, and that the rotational speed of the vane rotor increases when the capacity space of the vane chamber is decreased; that is, the vane rotor of each having a rotational speed in inverse proportion to the change in the capacity space of the vane chamber.

11. The driving method as claimed in claim 10, characterized in that a drive device of the at least one drive devices with variable suction/discharge amount is set an active drive device with variable suction/discharge amount and another drive device with variable suction/discharge amount is set a passive drive device with variable suction/discharge amount, the active drive device with variable suction/discharge amount and the passive drive device with variable suction/discharge amount being assembled to form a closed driving loop, the amount of the fluid in the closed loop being constant and unchanged, whereby when the capacity of the vane chamber of the active drive device with variable suction/discharge amount is enlarged, the capacity of the vane chamber of the passive drive device with variable suction/discharge amount is reversely minified, in the condition that a constant amount of fluid flows within the closed loop per unit time, the rotational speed of the vane rotor of the active drive device with variable suction/discharge amount being slowed down, while the rotational speed of the vane rotor of the passive drive device with variable suction/discharge amount being increased, the rotational speed of the vane rotor of the active drive device with variable suction/discharge amount being in inverse proportion to the rotational speed of the vane rotor of the passive drive device with variable suction/discharge amount, reversely, when the capacity of the vane chamber of the active drive device with variable suction/discharge amount is decreased, the capacity of the vane chamber of the passive drive device with variable suction/discharge amount being enlarged, in the condition that a constant amount of fluid flows within the closed loop per unit time, the rotational speed of the vane rotor of the active drive device with variable suction/discharge amount being increased, while the rotational speed of the vane rotor of the passive drive device with variable suction/discharge amount being slowed down, the rotational speed of the vane rotor of the active drive device with variable suction/discharge amount being also in inverse proportion to the rotational speed of the vane rotor of the passive drive device with variable suction/discharge amount.

12. The driving method as claimed in claim 11, characterized in that at least one of the movable wall member and the movable vane chamber sleeve in the active drive device with variable suction/discharge amount and the passive drive device with variable suction/discharge amount is forcedly pushed by an external force to make the movable wall member and the movable vane chamber sleeve of the active drive device with variable suction/discharge amount and the passive drive device with variable suction/discharge amount synchronously displace in the axial direction of the vane rotor, the synchronous displacement distance of the movable wall member and the movable vane chamber sleeve of the active drive device with variable suction/discharge amount being equal to the synchronous displacement distance of the movable wall member and the movable vane chamber sleeve of the passive drive device with variable suction/discharge amount.

13. The driving method as claimed in claim 11, characterized in that due to that the amount of the fluid in the closed loop being constant and unchanged, the capacity of the vane chamber of the active drive device with variable suction/discharge amount and the capacity of the vane chamber of the passive drive device with variable suction/discharge amount are synchronously changed and

increased/decreased in a complementary relationship, that is, when the movable wall member and the movable vane chamber sleeve of the active drive device with variable suction/discharge amount synchronously displace in the axial direction of the vane rotor toward the fixed wall member to minify the capacity of the vane chamber, the movable wall member and the movable vane chamber sleeve of the passive drive device with variable suction/discharge amount synchronously displace in the axial direction of the vane rotor away from the fixed wall member to enlarge the capacity of the vane chamber, the displacement distance of the movable wall member and the movable vane chamber sleeve of the active drive device with variable suction/discharge amount being equal to the displacement distance of the movable wall member and the movable vane chamber sleeve of the passive drive device with variable suction/discharge amount, reversely, when the movable wall member and the movable vane chamber sleeve of the active drive device with variable suction/discharge amount synchronously displace in the axial direction of the vane rotor away from the fixed wall member to enlarge the capacity of the vane chamber, the movable wall member and the movable vane chamber sleeve of the passive drive device with variable suction/discharge amount synchronously displacing in the axial direction of the vane rotor toward the fixed wall member to minify the capacity of the vane chamber, the displacement distance of the movable wall member and the movable vane chamber sleeve of the active drive device with variable suction/discharge amount being equal to the displacement distance of the movable wall member and the movable vane chamber sleeve of the passive drive device with variable suction/discharge amount.

14. A driving method for variable speed drive device, the variable speed drive device including at least one pumps with variable suction/discharge amount assembled to form an active drive device with variable suction/discharge amount, and at least one pumps with variable suction/discharge amount assembled to form a passive drive device with variable suction/discharge amount, and the active and the passive drive device with variable suction/discharge amount are then connected to together form an active and passive closed loop variable speed drive device; the pumps with variable suction/discharge amount respectively including a vane chamber body and a vane rotor, the vane chamber body internally having a vane chamber, which has a capacity space defined between a fixed wall member, a movable wall member and a movable vane chamber sleeve in the vane chamber body; the vane chamber being partitioned by an impeller of the vane rotor in the vane chamber into a plurality of eccentric vane chamber sections, a plurality of vanes being disposed on the impeller, and the number of the eccentric vane chamber sections being more than or equal to the number of the vanes; in the vane chamber, at any timing when the vane rotor continuously operates in the direction relative to the vane chamber body, one of two lateral sides of each vanes adjacent to another vane unchangeably being a suction side, while another one of the two lateral sides of each vane adjacent to another vane unchangeably being a discharge side, and the suction side and the discharge side respectively communicable suction discharge passages in communication with outer side of the pump; the fixed wall member being positioned in a fixed position in the vane chamber body; and the movable wall member and the movable vane chamber sleeve being displaceable in an axial direction of the vane rotor relative to the fixed wall member to increase or decrease and change the capacity space of the vane chamber, so as to form a pump capacity space with variable suction/discharge amount; the driving method being characterized in including the following steps: (1) causing the variable speed drive device to operate and allowing a difference value between a driving force of the active drive device with variable suction/discharge amount and a load resistance born by the passive drive device with variable suction/discharge amount; (2) as an effect of the difference value between the driving force and the load resistance, the movable wall members and the movable vane chamber sleeves of the active and the passive drive device with variable suction/discharge amount being subjected to a compressive push and a vacuum suction produced in the vane chamber, and accordingly being displace synchronously, causing the capacity space of the vane chamber of the active and the passive drive device with

variable suction/discharge amount to change and adjust their volume automatically under the effect of the difference value between the driving force and the load resistance; and (3) in the closed loop, when a balance of force is reached, the capacity spaces of the vane chambers of the active drive device and the passive drive device with variable suction/discharge amount being eventually automatically adjusted to a state in which the driving force of the active drive device with variable suction/discharge amount is equal to the load resistance of the passive drive device with variable suction/discharge amount, meanwhile, the volumes of the capacity spaces of the vane chambers and the rotational speed of the active and the passive drive device with variable suction/discharge amount also being automatically adjusted to be inversely proportional to each other during pump operation.